Nikhil Kadiyan

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Education

National Institute of Technology Delhi — Delhi, India
Bachelor of Technology (CGPA: 8.44) — 2020 – 2024

Experience

ThoughtSpot — Aug. 2025 – Present
Member of Technical Staff 3 | Python, Golang, Java — Hyderabad, India

- Reduced Prometheus metric cardinality by over **270,000 time series** in a **Java-based Kafka-driven Scheduler Service** by removing high-dimensionality labels (**client_id**, **org_id**) from request latency metrics, cutting monitoring infrastructure costs, improving dashboard query performance, and preventing metric storage from being overwhelmed at scale.
- Architected a **distributed, multi-tenant provisioning engine** using **Python, FastAPI, SQLAlchemy, Alembic,** and **Terraform on AWS** (EC2, S3) to automate isolated environment deployments across **3 global regions** (USA, Europe, Oceania), reducing onboarding deployment timeline from **weeks to under 2 hours** for **3k+ enterprise users**.
- Created **SpotterCode**, a custom **MCP server** handling **200 concurrent requests** via an async, multi-worker and horizontally scalable architecture and caching across 3 dedicated agents (embed, docs, API); enabling **5k+ enterprise users** to embed BI websites via natural language, **increasing embed adoption by 20%**.
- Engineered enterprise **SSO invitation and authentication flows** in **Java**, integrating with **Okta** and evaluating additional identity providers including **LinkedIn, GitHub, Discord, and Twitter**; resolved activation and identity-mismatch edge cases to maintain consistent user and invitation states.
- Improved **Sparkflow CSV upload latency by 67% in Java Callosum service**, reducing upload time from **30 seconds to 10 seconds** by eliminating a redundant database call through **BigQuery-aware request handling**.
- Created integrations for ThoughtSpot web plugins across **Google Slides, Sheets, Microsoft Excel, and PowerPoint**, enabling embedded analytics for **2k+ enterprise users**.
- Collaborated in building a **Salesforce LWC plugin** to embed ThoughtSpot dashboards directly within Salesforce CRM, reaching **300+ enterprise Salesforce admins**, a high-value segment at the intersection of two enterprise platforms.
- Developed **Java backend endpoints** for free-trial user creation and account reactivation, implementing **Apache Shiro token-based authentication** to securely manage user sessions and account lifecycle flows.
- Prototyped **Spotbuild**, an **AI-powered prompt-to-BI system** that auto-generates Liveboard answers and worksheets, pioneering an approach later adopted by the product's native AI features.

ThoughtSpot — Jul. 2024 – Jul. 2025
Member of Technical Staff 2 | Java, Golang, React, Typescript, GraphQL — Hyderabad, India

- Achieved a **90% reduction in homepage load time** by redesigning the **Java backend** and migrating the internal **Cassandra** impressions database to a new schema, replacing per-view lookups with a unified table storing total views per user.
- Introduced **org awareness** to the trending table of **Cassandra** database by creating a new table with altered schema, allowing each isolated org in the instance to have separate trending objects.
- Executed team-led **zero downtime data migration** of **50M+** rows across clusters in **Cassandra** database by implementing **a Java background migration service** that safely transferred data and seamlessly switched over to the new schema.
- Redesigned the user onboarding to highlight **AI-powered Spotter**, guiding users through creating connections, importing data, and asking questions via natural language, resulting in improved feature adoption by 30%.
- Addressed a **security vulnerability** in the share API by implementing a Redis-based rate limiter after penetration testing exposed a missing rate limit, effectively controlling API request rates and mitigating abuse risks.
- **Diagnosed and patched** a Mixpanel overtracking issue ahead of a renewal, **reducing test event volume by 98.5% (20M to 300K)** and **saving $30K**.
- Added frontend support for **monitor alerts in Slack**, enabling users to receive data updates directly in Slack without opening the BI tool, increasing notification click rate by 20%.

ThoughtSpot — Jun. 2023 – Jul. 2023 | Jan. 2024 – Jun. 2024
Software Engineering Intern | React, Typescript, Playwright, Jest — Hyderabad, India

- Migrated schema viewer from Angular to React by building 15+ React components and integrating GraphQL APIs, improving maintainability and reducing page load latency. Also rectified design issues in the legacy UI during migration.
- Implemented **filter and sort features** for worksheet tables and columns, collaborating with backend teams to extend APIs with **tags, authors, and database metadata**.
- Streamlined **50+ accessibility improvements** across the product's website using **React**.

Project Highlights

Event-Driven Order Processing Platform
- Designed a **distributed order-processing system** using **Java, Spring Boot, and Apache Kafka**, decoupling order creation, inventory reservation, and payment processing through asynchronous event-driven workflows.
- Implemented a **Transactional Outbox pattern** with **PostgreSQL** to prevent database updates and event publishing from becoming inconsistent, with idempotent consumers to safely handle duplicate message delivery.
- Built **Saga-based failure recovery** with retry policies, dead-letter topics, and compensating transactions to handle partial failures across distributed services without relying on a global database transaction.
- Configured **Kafka partitioning, consumer groups, and offset management** for scalable event processing, with **Docker Compose, Prometheus, and Grafana** for reproducible deployments and operational monitoring.

NEAT Twin-Stick Shooter Agent [Code]
- Built a **NEAT (NeuroEvolution of Augmenting Topologies)** agent from scratch in **C++**, evolving both neural-network topology and weights without ML libraries or gradient descent to play a twin-stick space shooter.
- Engineered a **41-feature real-time game-state representation** and self-growing neural architecture, improving score **52x** ($400 \rightarrow 20,800$) and survival time **18x** (15s $\rightarrow$ 278s) over 6,000 generations.
- Designed a **parallelized experiment framework** for population-size sweeps and repeated trials across CPU threads, reducing a 4-repeat, 6,000-generation, population-80 sweep to **12 minutes** wall-clock time.
- Implemented **deterministic seed-based reproducibility** across simulation, mutation, crossover, and selection, alongside automated checkpoint/GIF recording, genome visualization, and input-sensitivity analysis across 1,000+ evolved genomes.

Golang News Reader (UI + Terminal) [Code]
- Developed a full-stack RSS reader with a **Go backend**, **Preact frontend**, and interactive **terminal/web UI**.
- Implemented a **SQLite caching layer** and **REST API** in Go for feed management across **100+ feeds**.
- Supports full article extraction, TTS, podcast media enclosure, feed grouping, and persistent read/unread tracking.

Real Time Chatroom Application [Code+Screenshots]
- Built a multi-platform real-time chat system using **Next.js, Clerk authentication, and AWS S3 storage**.
- Supported **2GB media uploads** and real-time messaging with **Socket.IO**, deployed to **50+ test users**.

Technical Skills

Languages & Frameworks: TypeScript, Java, Golang, Python, C++, SQL, GraphQL, React, Spring Boot, Kafka, Next.js, Node.js, Flask
Tools & Cloud: AWS (EC2, S3, Lambda, CloudWatch), GCP, Terraform, Docker, Firebase, Git, Jest, Playwright

Achievements

- Solved **800+** Data Structures and Algorithm problems on **Leetcode** [Link], rated **1764** in contests.
- Achieved a highest rating of **1736 (Expert)** in **Codeforces** [Link].
- Published ICIC3S paper on **Plant Disease Identification Models** [IEEE Xplore].